

Datalink

Communication system for digital node-to-node communication

Can be wired or wireless.

Drone is attached to a wireless modem (here come the SIM cards), and acts like a node on the network

Latency?

Coverage?

Robustness?

[Datalink | EUROCONTROL](#)

European aviation datalink system is based on the Aeronautical Telecommunications Network (ATN)

VHF Digital Link Mode 2 (VDL Mode 2) -air ground datalink technology

All nodes on the system are connected to ATN routers

[Tactical Data Links | Collins Aerospace](#)

Rosetta system

LINK 16 based

Real time digital data exchange

MUOS

Mesh

Hf radios

[Link 16 - Wikipedia](#)

Link 16 is a TDMA-based secure, jam-resistant, high-speed digital data link that operates in the radio frequency band 960–1,215 MHz

Limited by the frequency range to LOS communications, which can be extended using satellite uplink and specialised protocols (TCP/IP)

[Jam proof radio communication : r/explainlikeimfive](#)

Frequency hopping - GPS clock or time table

Beam nulling - multiple antennas, forming an array. Turning specific antenna off, allowing detection only of a signal from the desired direction

CRPAs - spatial diversity; desired signals, and unwanted jamming signals, generally arrive from different directions.

In simple terms, you create a spatial filter, one that removes signals that arrive from particular directions, whilst letting through signals from other directions. To achieve this, rather than use a single antenna, we use an array of antenna elements.

Wide frequency spread

[How 5G Enables Real-Time Drone Operations | Anvil Labs](#)

Equipped with 5G modems, drones link to cell towers using network slicing.

Dedicated virtual networks, ensuring drone communications have their own bandwidth. Controls, telemetry and video.

Each drone connects securely through IMEI numbers and specialized SIM cards. Drones maintain a reliable connection using Quality of Service (QoS) protocols.

Continuous two-way data exchange. Commands from ground control stations are transmitted with incredibly low latency, giving remote pilots a level of responsiveness that closely mimics direct line-of-sight control.

To reduce latency for data analysis (not used for control) - edge computing. Instead of sending all data to distant servers, edge computing processes information locally at nodes near cell towers. Also compresses and filters information, reducing bandwidth

5G Drones: IoT Modules for Real-Time Drone Operations

LTE fallback

Useless without a proper regulatory framework

Only some of the functions are done locally, to better use computing power and reduce power usage

3GPP Release 16 5G NR Advanced Sub-6GHz module with LTE Cat 16 Fallback

- Up to 7.01 Gbps downlink
- Up to 1.25 Gbps uplink
- LTE Cat 20 fallback
- 4CC carrier aggregation
- Linux/OpenWrt
- Inbuilt GNSS
- USB 3.1
- PCIe 4.0 and PCIe 3.0
- Dual USIM
- UART, SPI, I2C, GPIO, and additional peripheral interfaces

Command and Control of Drones over 5G | Caballero E., Bettstetter C., Schupke D. a.o.

Using system-level simulations, we evaluate link availability, latency, and block error rate in an urban scenario with drones flying above ground users.

The requirements set by aviation safety authorities for beyond-visual-line-of-sight operations are only partially satisfied:

- link availability is adequate for basic operations (automatic flight, steering to waypoint) but insufficient for demanding ones such (stick control, control-over-camera missions)
- latency meets most needs but remains too high for direct stick control
- block error rate is generally too high (conflicting/unreliable commands)

These findings suggest that network configurations tailored to aviation safety requirements are needed.

Specific category flight

Gain pattern of the antenna specifically pointed down for ground users, not up

[Why don't drones use cellular for live transmission / RC? : r/dji](#)

Can be done DJI but very high latency. Regulatory issues

DJI 4G Cellular Dongle 2

[Is there a good way to do ultra-long-range FPV using phone signal? : r/drones](#)

High latency - more than 100ms. Would require custom hat for Arduino/Raspberry pie and code.

[5G Cellular Controlled Drone Boat - Long Range Test](#)

Cellular Datalink

[Elsight Halo: Unmanned - UAV/UGV - BVLOS Connectivity](#)

Video and Mavlink

Up to 4 sim cards

Usb to ardupilot

Ethernet to camera

[Using mobile cell phone hotspot with ardupilot flight controller](#)

Requires Raspberry Pie as bridge

STARLINK?

[How Starlink Has Changed the Game for Drones | Skyfront](#)

Starlink Mini mounted on the body - 1,1kilograms, 35-40 Watts, IP67, 100Mbps

Multilink architecture to protect against single point of failure

Gas-electric propulsion